

Bexley Crime Hotspot Project

Project 2 — Deliverable Summary

Tools: OpenAI Codex & Claude Pro | Language: Python | Data: Metropolitan Police Open Data

1. Project Description

This project investigates crime patterns in the London Borough of Bexley using three years of open Metropolitan Police street-level incident data covering March 2023 to February 2026, developed across two generative AI tools in a deliberate pipeline. The first notebook was produced with the aid of OpenAI Codex, which was used to pioneer the foundational data mining workflow: cleaning and exploring the dataset, engineering a binary hotspot target at the LSOA-month level, and training a tuned Random Forest classifier to predict which Lower Super Output Areas (LSOAs) are most likely to become crime hotspots in a given month, culminating in a safety score of 0 to 100 per area delivered through an interactive user interface. Building on this foundation, Claude Pro was then used to design and develop the second notebook, which extends the analysis by modelling Bexley's LSOAs as a spatial network using the `igraph` library — measuring whether hotspot status diffuses geographically to neighbouring areas over time, a phenomenon known in criminology as near-repeat victimisation. This two-tool approach demonstrates how different generative AI systems can be used in sequence, with the output of one serving as the conceptual and technical foundation for the other, producing an end-to-end data mining and graph analysis project hosted as a reproducible public repository on GitHub.

2. Deliverables Against Brief

Requirement	Status	Detail
Created with Claude Pro	Met	Claude Pro used to develop the <code>igraph</code> diffusion notebook (Notebook 2)
OpenAI Codex also used	Met	Codex used to pioneer the foundational hotspot prediction notebook (Notebook 1)
Interplay between data and programming	Met	Python used throughout: data cleaning, ML modelling, graph analysis and visualisation
Project description (paragraph)	Met	See Section 1 above
At least one dataset	Met	Metropolitan Police street-level CSV — 72 monthly files merged and filtered to Bexley

3. Dataset

Source

Metropolitan Police Service open crime data, downloaded from data.police.uk. Monthly street-level CSV files were collected for March 2023 to February 2026 and merged into a single Bexley-only dataset filtered to street-schema records containing full location information.

Dataset Details

Attribute	Value
File name	combined_metropolitan_data_bexley_street_only.csv
Time period	March 2023 – February 2026 (36 months)
Geographic scope	London Borough of Bexley
Spatial unit	LSOA (Lower Super Output Area)
Key fields	Month, Crime type, LSOA code, LSOA name, Latitude, Longitude
Hosted at	github.com/callumhudson0/Generative-AI-Final-Project
Licence	Open Government Licence v3.0

4. Technical Summary

Notebook 1 — Hotspot Prediction (01_hotspot_prediction.ipynb) | Generated with OpenAI Codex

- Cleaned and explored the merged Bexley street dataset across 36 months
- Engineered a binary Hotspot target using the 75th percentile of monthly LSOA crime counts
- Compared Logistic Regression, Decision Tree, and Random Forest classifiers
- Tuned the best model (Random Forest) using GridSearchCV
- Produced interactive heatmaps and LSOA-level crime visualisations using Folium
- Delivered a safety score (0–100) per LSOA via an interactive UI

Notebook 2 — Crime Diffusion Analysis (02_crime_diffusion_igraph.ipynb) | Generated with Claude Pro

- Built LSOA centroids from mean latitude and longitude of recorded incidents
- Constructed a spatial network using igraph: nodes = LSOAs, edges = areas within 1.2 km
- Measured crime diffusion rates at lag 1 and lag 2 months compared to a background baseline
- Calculated a lift multiplier to quantify the strength of the spatial diffusion effect
- Ranked LSOAs by a hub score (hotspot frequency x network degree) to identify diffusion hubs
- Visualised the full network and hub rankings on interactive Folium maps

5. GitHub Repository

All project files are hosted publicly and reproducibly at:

github.com/callumhudson0/Generative-AI-Final-Project

File / Folder	Description
notebooks/01_hotspot_prediction.ipynb	Main ML project notebook
notebooks/02_crime_diffusion_igraph.ipynb	igraph crime diffusion extension notebook
combined_metropolitan_data_bexley_street_only.csv	Bexley street-level crime dataset
requirements.txt	Python package dependencies
.gitignore	Git ignore rules for Python and Jupyter artefacts
README.md	Full project description and setup instructions